

Name: \_\_\_\_\_ Mark: \_\_\_\_\_ /60

**G10 Science - Class 3 Homework**

1. Complete the following table. [8 marks]

| Chemical Formula                    | Acid or Base? | Chemical Name |
|-------------------------------------|---------------|---------------|
| KOH (aq)                            |               |               |
| HNO <sub>3</sub> (aq)               |               |               |
| Ba(OH) <sub>2</sub> (aq)            |               |               |
| KHCO <sub>3</sub> (aq)              |               |               |
| H <sub>2</sub> SO <sub>4</sub> (aq) |               |               |
| LiOH (aq)                           |               |               |
| HI (aq)                             |               |               |
| H <sub>3</sub> PO <sub>4</sub> (aq) |               |               |

2. Consider these seven compounds: H
- <sub>3</sub>
- PO
- <sub>4</sub>
- (aq), HBr(aq), Fe(OH)
- <sub>3</sub>
- , H
- <sub>2</sub>
- SO
- <sub>4</sub>
- (aq), Ca(HCO
- <sub>3</sub>
- )
- <sub>2</sub>
- and KNO
- <sub>3</sub>
- . Predict which of these compounds will produce a solution with a pH: [6 marks]

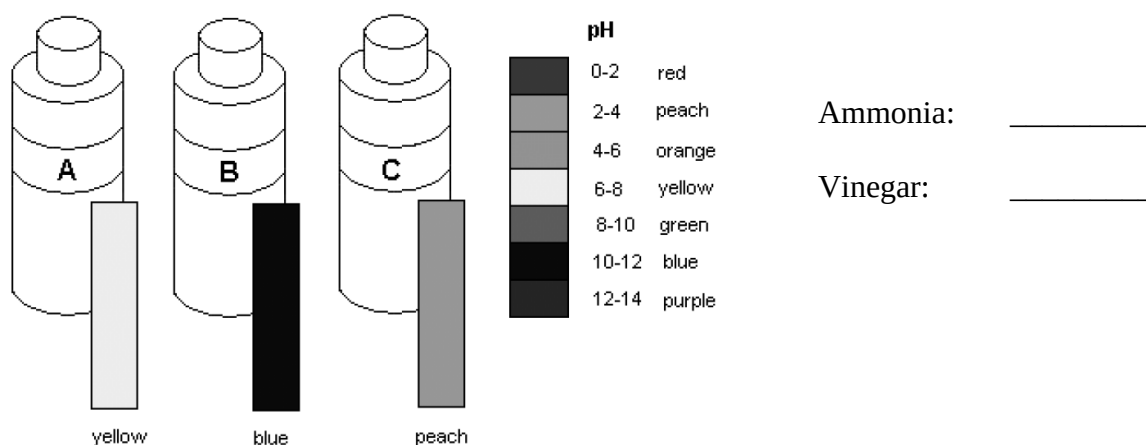
- a) Less than 7 \_\_\_\_\_
- b) Greater than 7 \_\_\_\_\_
- c) Equal to 7 \_\_\_\_\_

3. People who suffer from bulimia sometimes self-induce vomiting to prevent weight gain. Consider the properties of the chemical in your stomach. Why do the teeth of people with bulimia often appear worn or eroded? [2 marks]

4. A pink mixture of sodium hydroxide and phenolphthalein indicator was injected into a lemon. The lemon was cut in half. Explain why the inside of the lemon was not pink. **[2 marks]**

5. You find three unlabeled bottles of cleaning fluid in your house. You guess that one is vinegar and another one contains ammonia, but you do not know which is which.

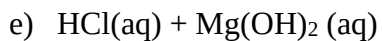
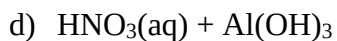
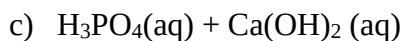
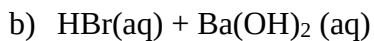
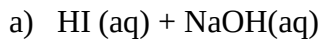
- a) You place universal indicator paper in each solution, with the following results. Which bottle corresponds to ammonia and which bottle corresponds to vinegar? **[2 marks]**



- b) Is the third solution's identity known? Provide your reasoning. **[2 marks]**
- c) Can the third solution be disposed of pouring it down the drain? Provide your reasoning. **[2 marks]**
6. You are asked to test three solutions: HCl(aq), NaCl(aq), and sugar water. Predict what you will observe in the following table. **[9 marks]**

| Solution    | Colour with Blue Litmus | Reaction with Mg(s) | Electrical Conductivity |
|-------------|-------------------------|---------------------|-------------------------|
| HCl(aq)     |                         |                     |                         |
| NaCl(aq)    |                         |                     |                         |
| Sugar water |                         |                     |                         |

7. Predict the products and balance the following neutralization reactions: **[10 marks]**



8. A substance with a pH of 2.3 is changed until its final pH is 6.7. Calculate how much more basic the final substance became. **[2 marks]**

9. Seafood often emits a fishy odour that is made of a class of organic compounds called amines. Amines are bases. Suggest why some seafood restaurants place a bowl of lemon water on the table to accompany a lobster or shellfish meal. **[3 marks]**

10. The hard parts of coral and shellfish are mostly made up of calcium carbonate. Evidence suggests that air pollution is causing the oceans to become slightly acidic.

- a) What kind of chemical reaction occurs when acids are in contact with calcium carbonate? **[1 mark]**
- b) Write the balanced chemical equation for the reaction of sulfuric acid with calcium carbonate. **[2 marks]**
- c) What effect would this reaction have on coral ecosystems? Explain. **[2 marks]**

11. You test the pH of your pool and find that the pH is 8.2.

- a) Is the pool water acidic or basic? **[1 mark]**
- b) At this pH, how is the water quality affected? **[1 mark]**
- c) How could you adjust the pH of your pool to be in the acceptable range? **[1 mark]**

12. Consider two neighbouring lakes in an isolated region of Ontario. One lake is on limestone ( $\text{CaCO}_3$ ) while the other is on granite. Both receive the same amount of acid precipitation.

- a) Explain why the limestone lake has a higher pH than the granite lake. **[2 marks]**
- b) Which lake is likely to have a healthier aquatic ecosystem? Explain your answer. **[2 marks]**